Linkset Usage Pattern: Large Linkset Split-up

Pattern Name

Large Linkset Split-up

Goal

The objective of this pattern is to provide a standardized mechanism for decomposing large linksets into manageable, cacheable, and specialized fragments. By utilizing the `rel="item"` and `rel="collection"` relations, providers can split extensive discovery maps—driven by large catalogs, complex sitemaps, or the aggregation of multiple RT patterns—without losing the structural integrity of the resource's "webmap."

Motivation

While [RFC 9264 linksets][RFC 9264] solve the "bloated HTTP header" problem, the linkset documents themselves can become a bottleneck at scale. Key drivers for this pattern include:

- Predictability and Performance: Massive linksets (e.g., for global catalogs) can impact parser performance and memory usage on constrained machine agents.
- Cache Optimization: Splitting links by their change frequency (e.g., static profile links in one file, dynamic provenance links in another) allows for more efficient HTTP caching strategies.
- Separation of Concerns: A provider may wish to group links by their functional role (e.g., one child linkset for the Content Negotiation Menu of RT-P03 and another for the Subsetting API anchors of RT-P05). Additionally various resources might be playing different roles in different patterns: so linksets could naturally reflect individual patterns and get recombined depending on specific resources.
- Sync Efficiency: In large-scale deployments like ResourceSync or LDES, breaking down updates into smaller chunks reduces the payload size for each synchronization event.

Relation to other patterns

This is a simple and stand-alone "architectural housekeeping" pattern. A direct application in fact of [RFC 6573], in particular to linksets [RFC 9264] themselves.

As such it simply serves as a reminder of a built-in engineering optimisation that might very well come in handy, precisely when eagerly adopting 'all these patterns' and its underlying Radical Transparency idea.

Encoding

Decomposition follows a hierarchical structure using the Item/Collection logic ([RFC 6573]):

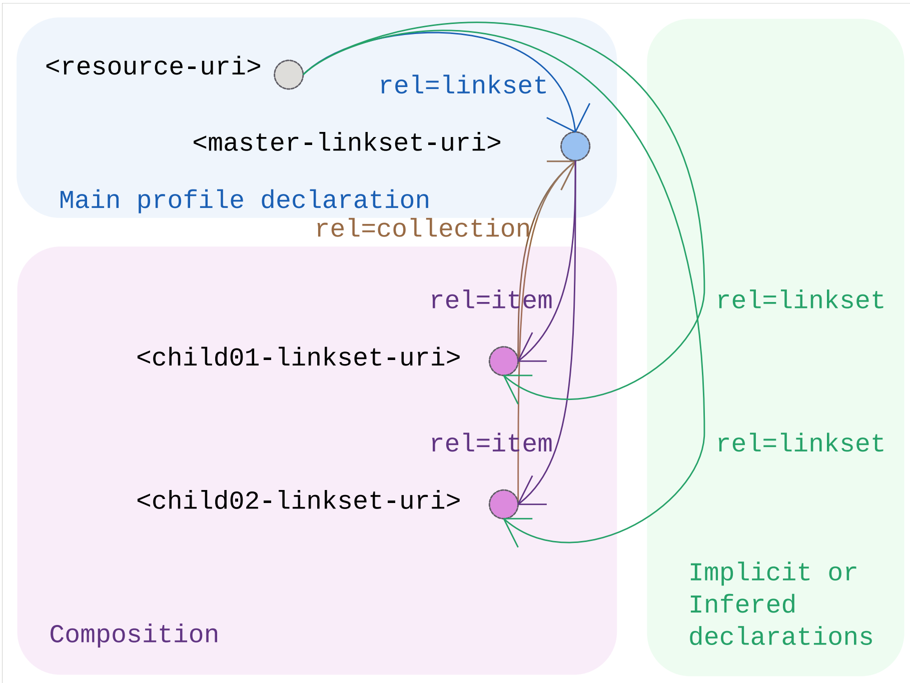
1. Identity Anchor: The resource itself points to a Master Linkset using rel="linkset".
2. Downlink (Decomposition): Within the Master Linkset, additional fragments are linked using rel="item".
3. Uplink (Context): Each child linkset SHOULD include a rel="collection" link back to the Master Linkset to maintain the discovery context.
4. Implicit Statement Scope: Statements within a child linkset remain autonomous. When a machine agent encounters an item relation in a linkset, it SHOULD recursively harvest the target to complete its understanding of the anchor resource.

Design Considerations: Target Attributes

While [RFC 8288] is open to any key/value target attributes, it is limiting in its own standard set of target attributes (e.g., type, title, media) in its 3.4 section, Radical Transparency encourages the use of extension attributes within linksets to guide machine agents. To optimize harvesting in RT-P08, providers SHOULD consider adding:

- `last-modified={iso 8601 datetime}`: To allow agents to skip hitting unchanged child linksets.
- `change= {created|updated|deleted}`: (Inspired by the changeType in ResourceSync) to signal the nature of the update within a split-up stream.

Sketch



Sketch of the linkset-usage-pattern for large-linksets

Link Relations Used

Relation Type	Specification	Technical Function
rel="linkset"	[RFC 9264]	Points the initial resource to the primary navigation map (the Master).
rel="item"	[RFC 6573]	Signals that the target is a constituent fragment of the linkset, triggering recursive discovery.
rel="collection"	[RFC 6573]	Points back from a fragment to its parent/master linkset to provide context.

Implementation Example: Pattern Weaving & Functional Split-up

To demonstrate RT-P08, we revisit some of the MarineInfo.org resources we mentioned in earlier patterns. As a provider adopts multiple Radical Transparency patterns, the number of link relations for a single resource grows: it needs to declare Conformity Profiles ([RT-P01]), expose a Content Negotiation Menu ([RT-P03]), and provide context for Subsetting APIs ([RT-P05]). Instead of serving one monolithic linkset, MarineInfo uses a Master Linkset to delegate discovery based on functional roles. This ensures that a harvester interested only in "provenance" doesn't have to parse the entire "variants menu."

1. The Master Linkset (institute-36.ls.json) The master linkset acts as the "Identity Anchor." It contains the bootstrap links and delegates the rest to specialized child linksets using `rel="item"`.

```json
{
  "linkset": [
    {
      "anchor": "https://marineinfo.org/id/institute/36",
      "item": [
        {
          "href": "https://marineinfo.org/id/institute/36/profiles.ls.json",
          "type": "application/linkset+json",
          "title": "Conformity & Profile Declarations (RT-P01/P02)"
        },
        {
          "href": "https://marineinfo.org/id/institute/36/variants.ls.json",
          "type": "application/linkset+json",
          "title": "Content Negotiation Menu (RT-P03)"
        },
        {
          "href": "https://marineinfo.org/id/institute/36/services.ls.json",
          "type": "application/linkset+json",
          "title": "API & Subsetting Context (RT-P05/P07)"
        }
      ]
    }
  ]
}
```

2. Specialized Child Linkset (Example: `profiles.ls.json`) This fragment focuses exclusively on the "What is this?" question, containing the relations from [RT-P01] and [RT-P02].

```json
{
  "linkset": [
    {
```

```
      "anchor": "https://marineinfo.org/id/institute/36",
      "collection": { "href": "https://marineinfo.org/id/institute/36.ls.json" },
      "profile": [
        { "href": "https://marineinfo.org/profiles/institute" },
        { "href": "https://schema.org/Organization" }
      ]
    }
  ]
}
```